

FX-aware Settlement Agent

An autonomous agent that settles cross-currency stablecoin payments at the best available **on-chain** rate on Arc.

DeFi Track

Agentic Economy Track

Live today

Real on-chain settlement + a public demo, priced by live Pyth feeds on Arc.

Built by

Jeong-hun Park — Korea-based quant (FX / basis / funding arbitrage).

PROBLEM

Cross-currency payments settle at the naive spot rate

Stablecoins drift from the currency they're pegged to. Nobody prices that gap before paying — so payers quietly overpay, every time.

30–50 bps

typical EURC deviation from its euro peg, observed live on Arc via Pyth.

~\$3,000

left on the table on a single €1M B2B settlement at 30 bps.

- Remittances, marketplace payouts, global payroll — all convert at spot.
- As **AI agents** start settling autonomously, this leakage scales with them.

SOLUTION

The agent prices the route before it pays

- **Reads three live Pyth feeds on Arc** — EUR/USD (FX), EURC/USD, USDC/USD — with confidence + staleness on every read.
- **Computes the basis:** how far EURC trades from its own euro peg.
- **Routes the cheaper way:** buy discounted EURC on-chain, or settle direct in USDC when there's no edge.
- **Settles in USDC** via a Circle Developer-Controlled Wallet — server-side signing, no private keys in the app — with a memo for reconciliation.
- **Charges a sub-cent nanopayment** per settlement for the routing service.

Safety by design: if the oracle is stale or its confidence is wide, the agent refuses the FX bet and settles safely in USDC.

IT'S REAL, NOT A MOCK

Live on Arc testnet

Real on-chain settlement

1 USDC, FX-aware route, memo attached, signed server-side by a Circle Dev-Controlled Wallet.

0xbef17f3513914f502012c81fcb4e7252464e6306b8f8a6e5238f9d302691234f

Live pricing

EUR/USD **1.1455** from Pyth on Arc · EURC $\approx$ **50 bps** below its peg → route **onchain_swap**.

Public demo

`fx-aware-settlement-agent.onrender.com/present`

Plans with live on-chain data; execution stays key-gated.

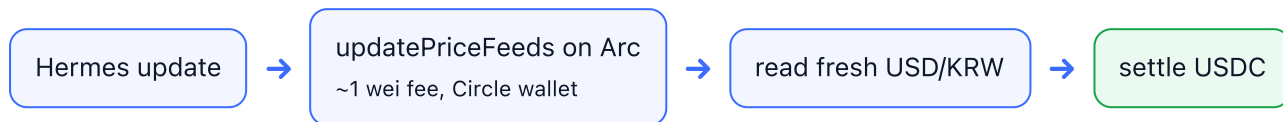
Open source

MIT, ~600 lines, zero-dependency Arc reader + Circle wallet helpers others can lift.

FLAGSHIP CORRIDOR

KRW1 / USDC — won settlement at an on-chain rate

USD/KRW isn't kept warm on Arc, so the agent refreshes it itself — Pyth's pull model, exactly as intended.



₩1,000,000

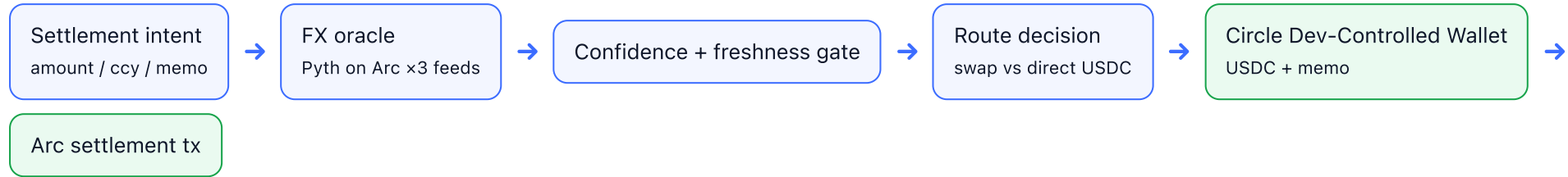
settles for **672 USDC** at the on-chain rate the agent just published.

~10 USDC

saved vs a typical ~1.5% bank FX spread — on one payment.

ARCHITECTURE

Intent → price → gate → route → settle



Circle / Arc stack

USDC (native gas) · Developer-Controlled Wallets ·
Nanopayments · Pyth oracle · transaction memos

Why Arc

USDC-denominated fees, deterministic sub-second finality,
stablecoin-native FX rails

From hackathon to something I keep running

- **Depth-aware confidence** — size the trade against pool depth and expected slippage, not just price confidence.
- **More corridors** — GBP and JPY feeds are already live on Arc; StableFX partner stablecoins as they land.
- **KRW1 end-to-end** — settle into the won stablecoin itself once it's on Arc, closing the corridor.
- **Agent-to-agent** — the agent as a paid routing service other agents call, billed per settlement.

Thesis: stablecoin FX dislocations are real, Arc makes them programmable, and an agent that prices the route before it pays turns that into savings — autonomously, per payment.